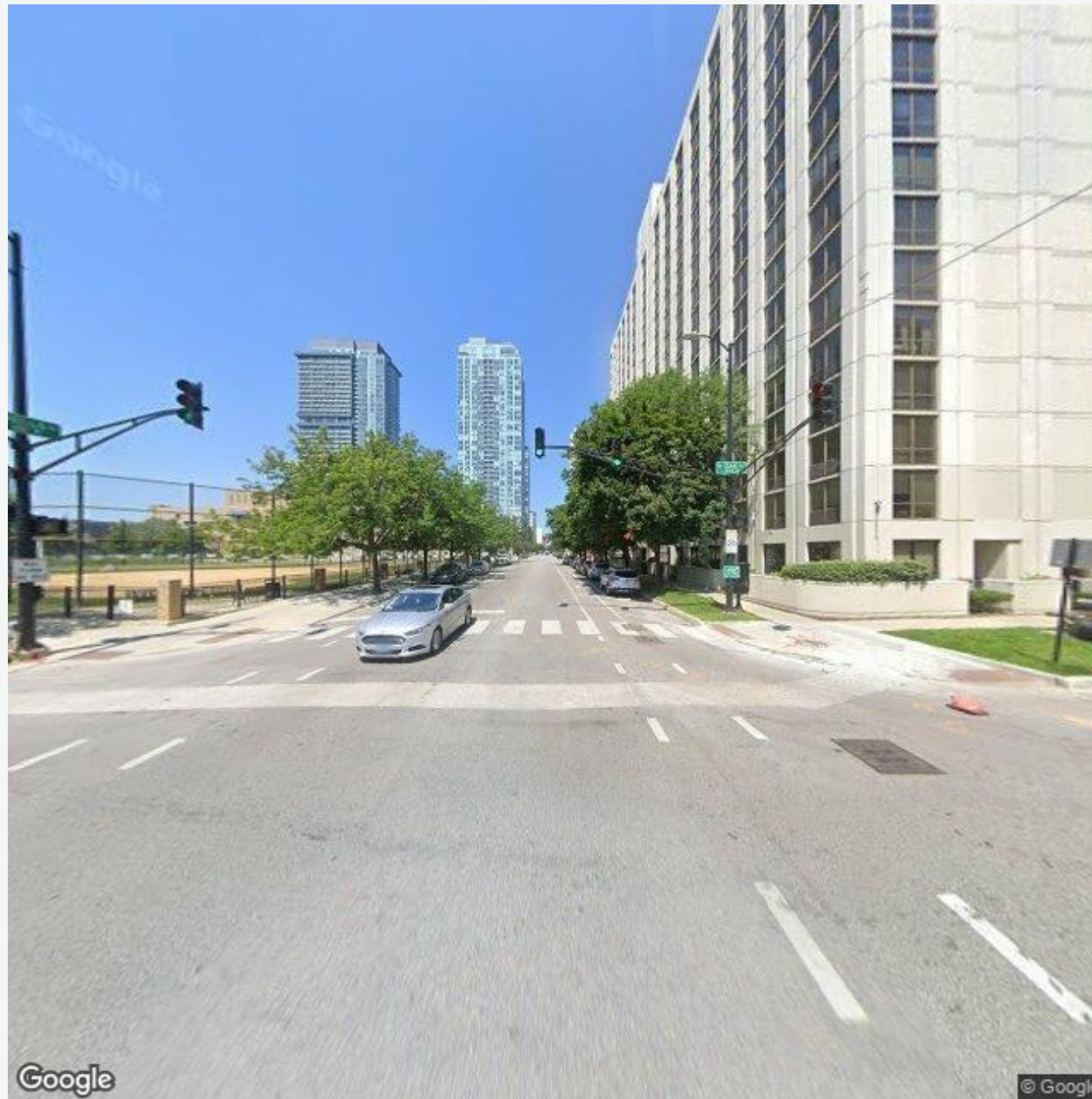
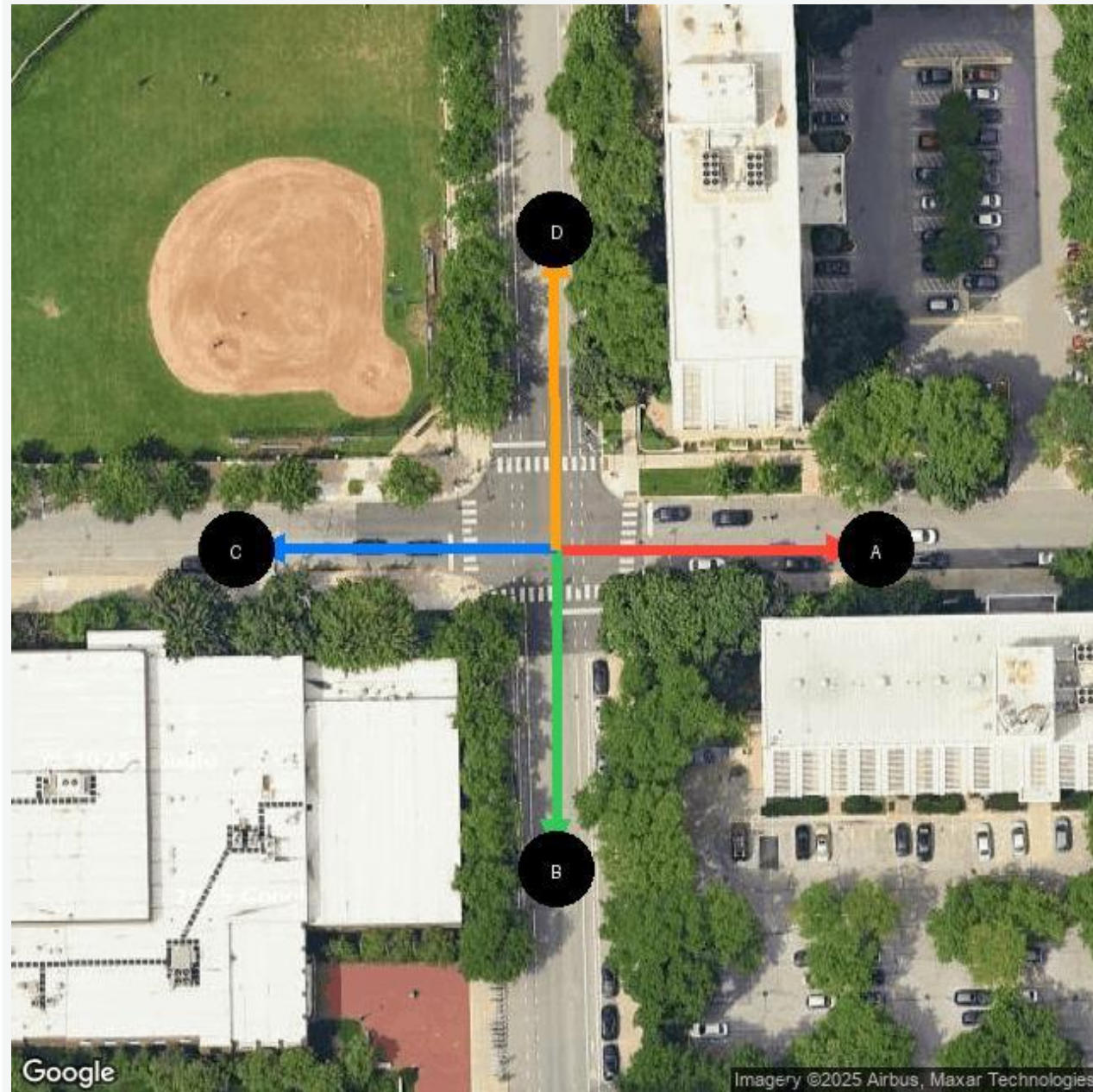


m2sv: A Scalable Benchmark for Map-to-Street-View Spatial Reasoning

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Combining Theory and Benchmarks (CTB) Workshop @ ICML 2026 · **non-archival**
github.com/YosubShin/m2sv · live demo: m2sv.yosubshin.com

The Task

Given a **north-up overhead map** with labeled candidate directions and a **Street View photo** taken at the same intersection, infer *which labeled direction the camera was facing*. The camera sits at the intersection center; letters point outward along each road.



Solve by aligning stable map geometry (roads, parcels, landmarks) with the ground-level view — not transient cues.

Why it is hard

- **Allocentric** ↔ **egocentric** alignment across two very different viewpoints.
- **Temporal mismatch**: map and photo may be captured years apart.
- **Geometric ambiguity**: symmetric intersections admit multiple plausible headings and demand fine-grained cue escalation.

Benchmark & Data (open)

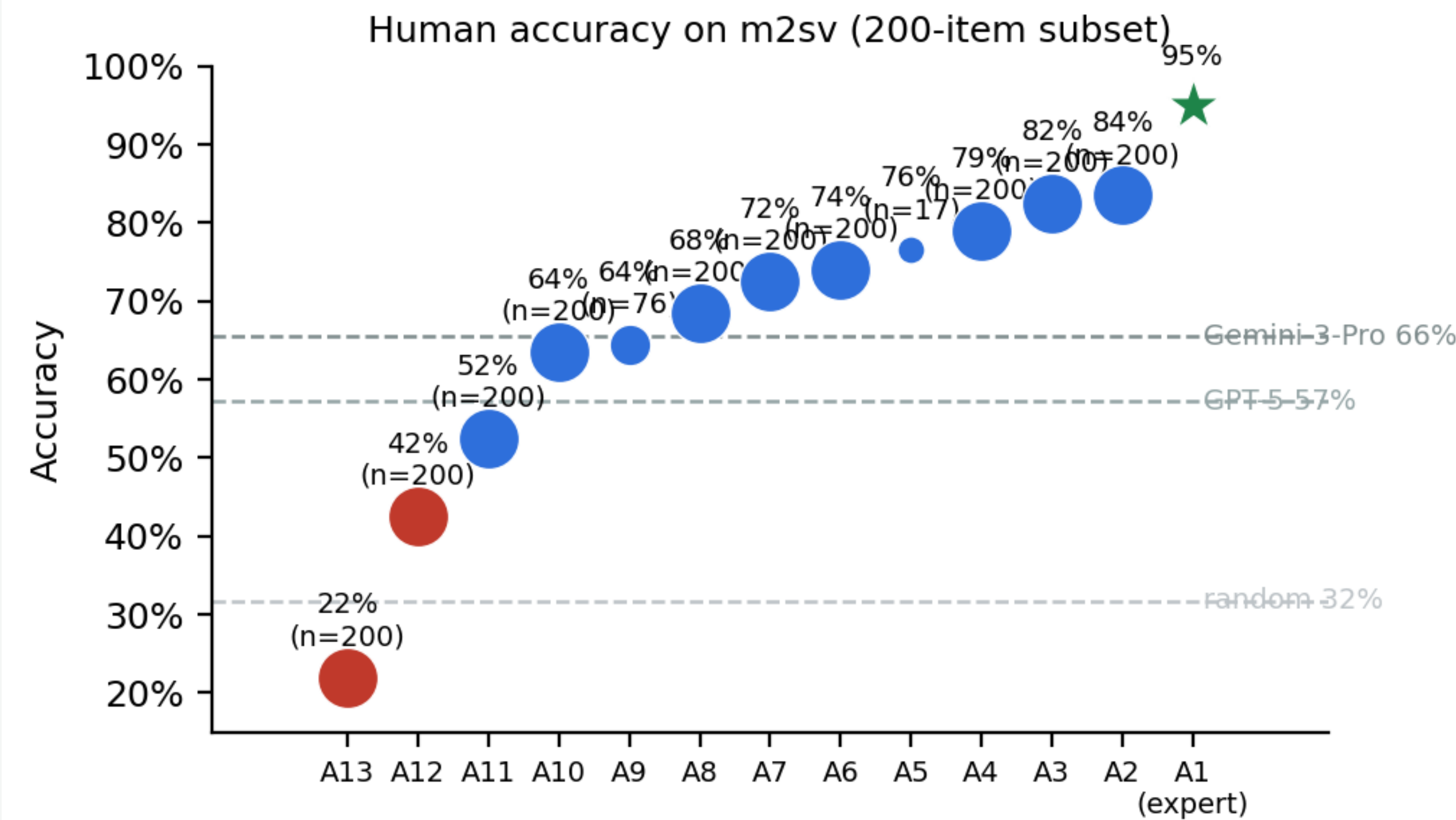
- **m2sv-20k** — 20,000 real intersections (10k train / 10k validation) across **32 cities**, multiple continents, with controlled candidate ambiguity.
- **m2sv-sft-11k** — 11k curated structured reasoning traces for supervised fine-tuning.
- Released as **blueprints** (coordinates + azimuths) plus rendering code to respect map licensing; all eval/training code public.

Zero-shot: humans clear the best VLM

Model	N	Acc. (%)
Gemini-3-Pro	1k	65.2 ± 3.0
GPT-5	1k	57.2 ± 3.1
Gemini-2.5-Pro	1k	47.2 ± 3.1
Qwen3-VL-235B (Think)	1k	42.7 ± 3.1
Random baseline	1k	31.4 ± 2.9
Human (engaged, n=10)	200	71.7 ± 9.1
Human (expert)	200	95.0

New multi-annotator study (12 annotators, 10 finished all 200) replaces the single-annotator pilot. Engaged = Cohen's $\kappa_{\text{expert}} \geq 0.3$; two low-agreement outliers excluded by *agreement*, not accuracy.

Human baseline & inter-rater agreement



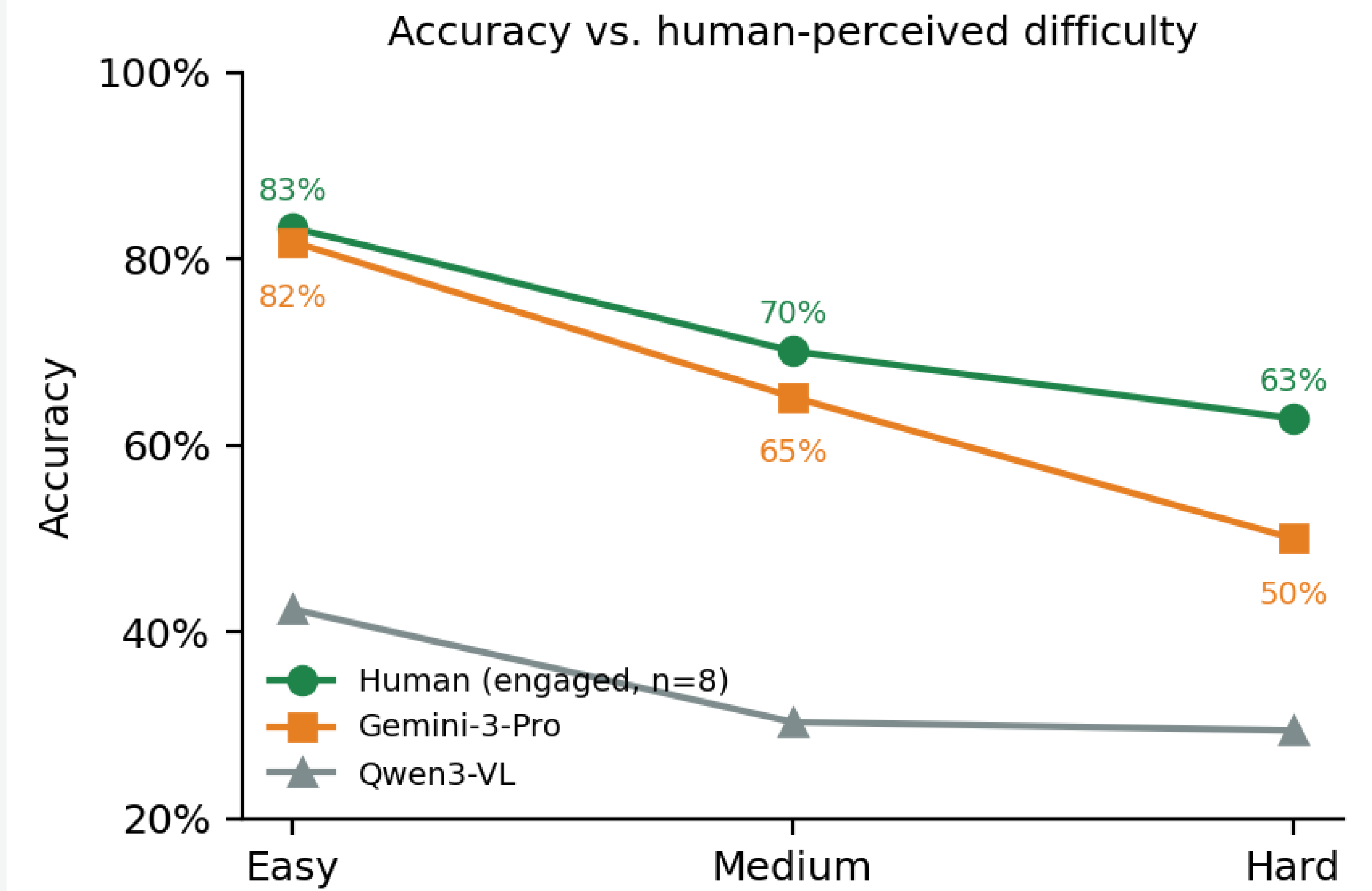
Attentive humans converge (median pairwise $\kappa = 0.48$, up to 0.76); Gemini-3-Pro sits *below* the human consensus (median $\kappa = 0.38$), Qwen3-VL near chance ($\kappa \approx 0.05$). The gap shows up in *agreement structure*, not only accuracy.

Adaptation helps, but transfer is limited

Model	Training	Acc. (%)
Qwen3-VL-8B Base		34.3 ± 0.9
Qwen3-VL-8B SFT		39.8 ± 1.0
Qwen3-VL-8B SFT+RL		43.9 ± 1.0

SFT + RL gains ~10 points on m2sv, but cross-benchmark transfer is asymmetric and weak — fine-tuning induces *benchmark-specific* policies, not transferable spatial skill.

Difficulty: the gap widens where it matters



Difficulty bins from **human response time**; accuracy from *separate* annotators (de-circularized). Easy→Hard: humans 83→63%, Gemini 82→50%, Qwen 42→29%. Gemini nearly matches humans on easy items but trails by **13 pts** on hard ones.

Road-azimuth symmetry. We define a structural difficulty metric $S = \frac{\max \text{ angular gap}}{\min \text{ angular gap}}$. Stratifying 3-option items by S , *symmetric* intersections are hardest for all models.

How models fail

- **Ego↔allo confusion** — left/right inversions at 4-way junctions.
- **Unreliable cues** — roof color, shadows, blur drive confident wrong eliminations.
- **Landmark misbinding / false projection** — attaching distant or absent landmarks to the intersection.
- **Symmetry traps** — fail to escalate to lane counts, markings, curb geometry that resolve ambiguity.

Common thread: models apply *uniform* reasoning across difficulty; humans escalate effort on ambiguous cases.

Takeaways

- m2sv isolates a core, scalable **map↔street-view alignment** primitive that frontier VLMs have *not* solved.
- Humans beat the best model and **agree with each other**; models lie outside that consensus.
- Difficulty is real and structural — failures concentrate on *symmetric* geometry.

Code/data: github.com/YosubShin/m2sv
Try it: m2sv.yosubshin.com